

ENTERPRISE UNIVERSITY EVENT MANAGEMENT & DYNAMIC TICKETING SYSTEM

Institutional Architecture, Deterministic State Governance & Clearance Specification

DOCUMENT VERSION	CLASSIFICATION	TARGET PLATFORM	GOVERNANCE BODY
v2.5.0-PROD	Official Technical Standard[cite: 1]	Campus ERP & Gate Terminal[cite: 1]	Office of the Registrar & VPAA[cite: 1]

1. EXECUTIVE SUMMARY & LIFECYCLE ARCHITECTURE

The Enterprise University Event Management and Dynamic Ticketing System governs the end-to-end lifecycle of university-wide events, co-curricular assemblies, and departmental symposia[cite: 1]. It unifies administrative event governance, pre-verified student eligibility enforcement, real-time gate admission via encrypted optical QR tokens, and tamper-resistant registrar accreditation clearance[cite: 1].

Pre-Provisioned Registrar Store[cite: 1]

Enforces student identity verification via immutable Student ID, University Branch, Department, and Degree Course records[cite: 1].

Live Gate Terminal (< 2.0s Validation)[cite: 1]

Real-time optical scanner supporting 4 instant verification states: Access Granted, Duplicate Alert, Revoked, and Invalid Record[cite: 1].

Targeted Catalog Discovery[cite: 1]

Dynamic event display filtered across 4 cohort restriction vectors: Campus Branch, Department, Degree Program, and Year Level[cite: 1].

The Dynamic Cancellation Bounce[cite: 1]

Automatically reopens fully-booked events upon student self-cancellation strictly within the open registration window[cite: 1].

2. INSTITUTIONAL ROLE HIERARCHY & GOVERNANCE MATRIX

System access is consolidated strictly into two authenticated user tiers, with administrative users directly executing event planning, live gate ingress scanning, and post-event clearance[cite: 1]:

Institutional Role	Permitted Operational Scope	Access Constraints & Authentication
Institutional Administrator [cite: 1]	Full event creation, core logistics management, manual early lock execution, event cancellation, student profile pre-provisioning, manual ticket resends, optical QR check-in scanning (< 2.0s ingestion), dead-battery manual override execution, supervisor scan rollbacks, velocity curve auditing, and official accreditation clearance generation[cite: 1].	MFA-protected Administrative Portal; master institution credentials[cite: 1].
Student Attendee [cite: 1]	Discovery of targeted event feeds, one-click reservation confirmation, input of section and year verification, QR ticket dashboard access, and self-service cancellation strictly during the open registration window[cite: 1].	Verified university credentials with mandatory first-login password update[cite: 1].

3. STUDENT PROFILE PRE-PROVISIONING & ACCOUNT SECURITY

Anti-Fraud Identity Mandate: Open student self-registration is strictly prohibited[cite: 1]. All student user accounts are batch-ingested by institutional administrators directly from verified registrar master records to eliminate fictitious identities and duplicate seat reservations[cite: 1].

- **Pre-Provisioned Master Fields:** Student Identification Number (Primary Key), Legal Last Name, Legal First Name, Gender, University Branch, College/Department, Degree Program/Course, Institutional Email (@university.edu), and Temporary Placeholder Password[cite: 1].
- **Forced Credential Update:** Upon initial login, students are immediately routed to a mandatory password change screen[cite: 1]. Access to event browsing, registration, and dashboard features is locked until new credentials are confirmed[cite: 1].
- **Profile Immutability:** All registrar-provisioned placement fields remain read-only on the student dashboard[cite: 1]. Discrepancies must be escalated to the university registrar to prevent unauthorized cohort shifts[cite: 1].

4. EVENT CONFIGURATION & TARGET AUDIENCE MATRIX

Administrators configure event logistics and targeting parameters through structured operational forms[cite: 1]:

Configuration Parameter	Technical & Operational Definition
Core Logistics[cite: 1]	Event Title, Rich-Text Description, Physical Venue Name, Room/Hall Number, and Total Seating Capacity[cite: 1].
Temporal Boundaries[cite: 1]	Event Date, Start Time, and End Time, combined with Registration Window Start Date/Time and Registration Window End Date/Time[cite: 1].
Institutional Identity[cite: 1]	Lead Hosting Department/Organization, Co-Sponsors, and Official University Partner Badges[cite: 1].
Target Audience Matrix[cite: 1]	Multi-select restriction rules configured across four discrete vectors: University Branch, College/Department, Degree Program/Course, and Year Level[cite: 1].

Target Audience Filtering Logic: Events configured with restrictions are visible only to students whose pre-provisioned master profile matches every active rule[cite: 1]. Students outside designated criteria cannot view or register for the event[cite: 1].

5. STUDENT REGISTRATION & DYNAMIC CANCELLATION ENGINE

5.1 Booking Workflow & 1-Student/1-Ticket Invariant

- Targeted Catalog Discovery:** Eligible students view tailored event cards displaying schedules, venue locations, host badges, registration countdown timers, and live seating availability[cite: 1].
- Verification & Submission:** Selecting an event opens a confirmation dialogue displaying read-only identity fields (Student ID, Full Name, Gender, Branch, Department, and Course)[cite: 1]. The student verifies their Year Level and enters their assigned Section[cite: 1].
- One-Ticket Rule:** The database enforces an absolute constraint of one active ticket per student per event at the transactional boundary[cite: 1]. Duplicate booking attempts are blocked[cite: 1].
- Issuance & QR Delivery:** Upon confirmation, the live seat count increments, the event appears on the student dashboard, and an encrypted QR ticket is dispatched to their verified institutional email[cite: 1].

5.2 Window-Tethered Cancellation & "The Cancellation Bounce"

- The Cancellation Bounce:** When an event reaches maximum capacity (e.g., 200/200), its status switches to FULLY_BOOKED and registration locks[cite: 1]. If a student cancels while the registration window remains open, the count drops to 199/200, and status immediately flips back to OPEN, unlocking public registration without administrator intervention[cite: 1].
- Registration Window Cutoff:** Students can cancel their registration directly from their personal dashboard strictly while the registration window remains active (prior to Registration End Date/Time)[cite: 1].
- Permanent Operational Roster Freeze:** As soon as the registration window closes, the self-cancellation button is permanently hidden and disabled on all student dashboards[cite: 1]. Final rosters freeze to allow administrators to plan venue resources and identify true no-shows[cite: 1].
- Scan Lockout:** Once a student's QR ticket is scanned at the venue entrance, its status updates to CHECKED_IN, permanently revoking self-cancellation privileges regardless of remaining window time[cite: 1].

6. DETERMINISTIC 5-TIER EVENT STATE MACHINE

System operations, registration buttons, and cancellation controls are governed by a deterministic priority hierarchy[cite: 1]. Higher-tier administrative overrides supersede lower-tier automatic state conditions[cite: 1]:

System State	Registration?	Self-Cancel?	Dynamic Bounce?	Trigger Conditions & Precedence Rules
CANCELLED (TIER 1) [cite: 1]	No[cite: 1]	No[cite: 1]	No[cite: 1]	Admin cancels event; mandatory reason logged; tickets invalidated; batch notification emails sent[cite: 1].

System State	Registration?	Self-Cancel?	Dynamic Bounce?	Trigger Conditions & Precedence Rules
MANUALLY LOCKED (TIER 2) [cite: 1]	No[cite: 1]	No[cite: 1]	No[cite: 1]	Admin triggers "Close Registration Early"[cite: 1]. Manual kill switch overrides capacity rules and prevents reopening[cite: 1].
REGISTRATION CLOSED (TIER 3) [cite: 1]	No[cite: 1]	No[cite: 1]	No[cite: 1]	Current Time >= Registration End Timestamp[cite: 1]. Roster locks; self-service cancellation permanently disabled[cite: 1].
FULLY BOOKED (TIER 4) [cite: 1]	No[cite: 1]	Yes[cite: 1]	Yes[cite: 1]	Confirmed registrations reach 100% capacity during active window[cite: 1]. A cancellation dynamically reopens the event to OPEN[cite: 1].
OPEN (TIER 5) [cite: 1]	Yes[cite: 1]	Yes[cite: 1]	N/A[cite: 1]	Active registration window, available capacity, and no active administrative overrides[cite: 1].

7. ADMIN PHASE 1: PLANNING & PRE-REGISTRATION MONITORING

Following event configuration, administrators monitor capacity and manage rosters through a real-time command dashboard[cite: 1]:

7.1 Event Overview & Live Controls

- **Dynamic Header & Status Badge:** Displays real-time operational status (OPEN, FULLY_BOOKED, REGISTRATION_CLOSED, MANUALLY_LOCKED, or CANCELLED)[cite: 1].
- **Core Logistics Card:** Shows schedule, physical venue, hall capacity, and an active registration deadline countdown timer[cite: 1].
- **Capacity Progress Meter:** Live visual gauge tracking reserved slots against venue limits (e.g., *210 / 250 Seats Taken (84%)*)[cite: 1]. Public registration locks automatically upon reaching 100% capacity[cite: 1].
- **Administrative Action Controls:**
 - *Edit Event:* Updates schedule, venue, or host details, automatically sending notification emails to all registered students[cite: 1].
 - *Close Registration Early:* Engages the MANUALLY_LOCKED state override to freeze registrations and prevent reopening[cite: 1].
 - *Cancel Event:* Prompts for a mandatory cancellation reason, voids all issued tickets, and dispatches batch cancellation emails[cite: 1].
 - *Resend All Ticket Reminders:* Re-transmits QR code tickets to all confirmed attendees[cite: 1].

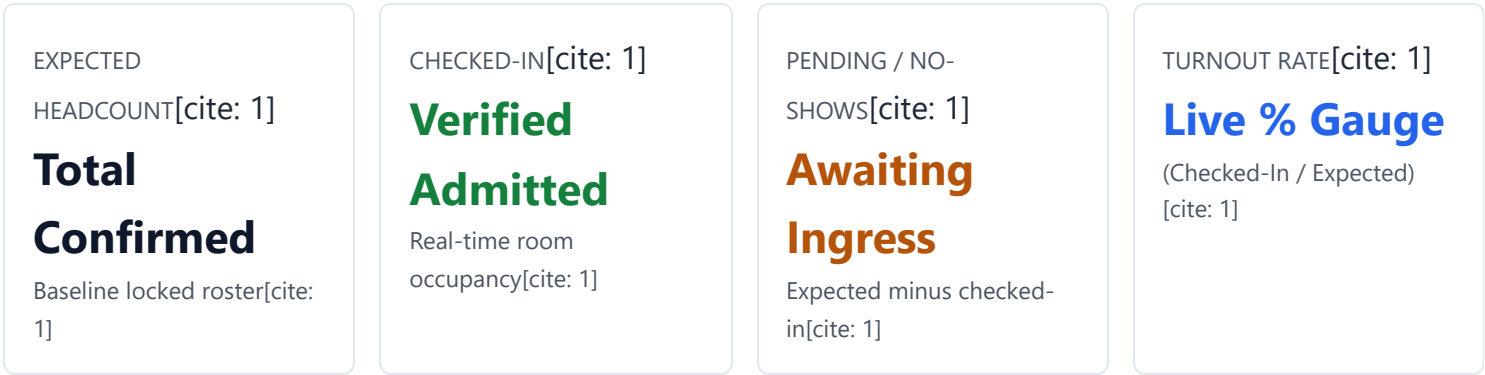
7.2 Pre-Registration Roster Table

- **Attendee Data Columns:** Student Number, Full Legal Name, University Branch, College/Department, Degree Program/Course, Year Level, Section, Registration Timestamp, and Ticket Delivery Status (Sent or Failed)[cite: 1].
- **Search & Filter Tools:** Search by Name or Student ID; filter records by Campus Branch, Department, Year Level, or Section[cite: 1].
- **Roster Operations:** Single-student ticket resend utility for attendees reporting lost or filtered confirmation emails[cite: 1].

8. ADMIN PHASE 2: LIVE GATE EXECUTION & QR TERMINAL

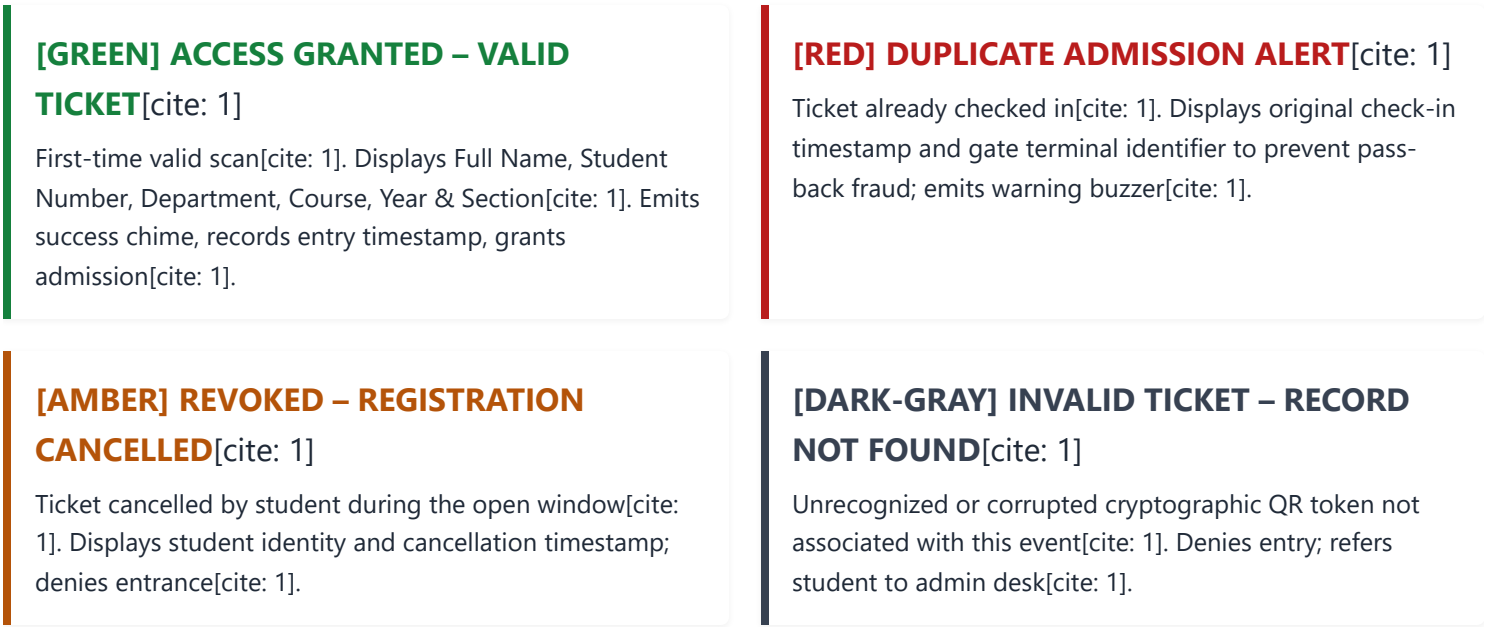
During the event, administrators operate the unified live gate terminal designed for high-throughput scanning (< 2.0s per student)[cite: 1]:

8.1 Live Command Headcount KPI Deck



Streaming Audit Feed: Chronological live audit log streaming attendee names, student IDs, gate terminal ID, and exact entry timestamps.

8.2 Optical QR Scanner & 4-State Verification Terminal



8.3 Operational Fallbacks & Gate Exceptions

- Manual Lookup & Override:** Search input by Student Number or Last Name to admit attendees with dead phone batteries or damaged screens. Manual admissions require a mandatory reason flag (e.g., *Dead Battery, Unsent Email*) for audit compliance.
- Dual-Filter Attendance Roster:** Live split-table interface separating registered students into *Present* and *Pending Arrival* tabs with real-time search.
- Supervisor Scan Rollback:** Admin capability to reverse accidental scans, returning the record to *Pending Arrival* and adjusting headcount metrics, while keeping student self-service cancellation locked.

9. ADMIN PHASE 3: POST-EVENT ANALYTICS & CLEARANCE HUB

9.1 Performance Analytics & Velocity Curves

- **Final Turnout Metrics:** Final summary comparing total seating capacity, confirmed pre-registrations, verified attendees present, final turnout percentage, and net no-shows[cite: 1].
- **Audience Purity Percentage:** Percentage of attendees who strictly matched target audience filter criteria versus authorized manual exceptions[cite: 1].
- **Demographic Breakdown Visualizations:** Breakdown of turnout across Academic Departments, Year Levels, Sections, and Campus Branches[cite: 1].
- **Arrival Velocity Curve:** Time-series distribution tracking check-in volume across 10- or 15-minute intervals to highlight peak ingress windows and door bottlenecks[cite: 1].

9.2 Master Audit & Clearance Export Deliverables

1. **Master Raw Dataset (.xlsx):** Unaggregated operational spreadsheet containing Student ID, Full Legal Name, University Branch, College/Department, Degree Program, Year Level, Section, Registration Timestamp, Ingress Check-In Timestamp, Gate Terminal ID, and Verification Mode (Optical Scan vs. Manual Override with audit notes) for registrar ingestion[cite: 1].
2. **Official Certified Attendance PDF:** Tamper-evident institutional clearance certificate formatted below[cite: 1]:

OFFICIAL UNIVERSITY ACCREDITATION & ATTENDANCE CLEARANCE CERTIFICATE

Office of the University Registrar • Division of Academic Affairs • Published via Verified Campus Audit Ledger[cite: 1]

Event Title: Annual Computer Science Symposium 2026[cite: 1]

Event Date: September 26, 2026[cite: 1]

Accreditation Ref: CS-2026-SYM-0982[cite: 1]

Hosting Entity: Department of Computer Studies[cite: 1]

Verified Present: 212 Attendees (84.8% Turnout)

Seating Capacity: 250 Seats[cite: 1]

Student ID	Full Legal Name	Department & Program	Yr/Sec	Check-In Time	Ingress Verification Mode
2024-10821[cite: 1]	Doe, Jane Alyssa[cite: 1]	Computer Science / BSCS[cite: 1]	3-A[cite: 1]	12:48:12 PM[cite: 1]	Gate 01 (Optical Scanner)[cite: 1]

Student ID	Full Legal Name	Department & Program	Yr/Sec	Check-In Time	Ingress Verification Mode
2024-11409[cite: 1]	Smith, John Eric[cite: 1]	Information Tech / BSIT[cite: 1]	3-B[cite: 1]	12:51:04 PM[cite: 1]	Gate 02 (Optical Scanner)[cite: 1]
2024-08992[cite: 1]	Vance, Robert M.[cite: 1]	Computer Science / BSCS[cite: 1]	4-A[cite: 1]	12:55:40 PM[cite: 1]	Gate 01 (Manual Override: Dead Battery) [cite: 1]

Prof. Elizabeth Sterling, M.Sc.

Lead Event Coordinator, Department of Computer Studies

Date Signed: September 26, 2026[cite: 1]

Dr. Arthur Vance, Ph.D.

Dean, College of Computer Studies
Date Signed: September 26, 2026[cite: 1]

Institutional Cryptographic Verification Seal: SHA256: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca49599167852b855[cite: 1]

10. CONCEPTUAL DATA MODEL & ENTITY RELATIONSHIPS

The relational database schema is structured around three core operational entities[cite: 1]:

Conceptual Entity	Key Functional Attributes	Relational Invariants & Integrity Constraints
STUDENTS <i>(Master Ingestion)</i> [cite: 1]	Student_ID (PK), First_Name, Last_Name, Gender, Branch_Name, Department_Name, Course_Name, University_Email, Password_Hash, Password_Change_Required[cite: 1].	Pre-provisioned by administrators[cite: 1]. Student_ID and University_Email are globally unique[cite: 1]. Profile placement fields remain strictly immutable to the student[cite: 1].
EVENTS <i>(Event Configuration)</i> [cite: 1]	Event_ID (PK), Title, Description, Venue_Location, Max_Capacity, Current_Registrations, Event_Start, Event_End, Reg_Window_Start, Reg_Window_End, Target_Filters, Status, Manual_Lock_Flag[cite: 1].	Event status is governed by the 5-tier state machine[cite: 1]. Dynamic reopening ("The Cancellation Bounce") triggers only when Manual_Lock_Flag == FALSE and current time is within registration boundaries[cite: 1].
TICKETS / ROSTER <i>(Junction Entity)</i> [cite: 1]	Ticket_ID (PK), Event_ID (FK), Student_ID (FK), Year_Confirmed, Section_Confirmed, Ticket_Token, Status (ACTIVE, CHECKED_IN, REVOKED), Reg_Timestamp, CheckIn_Timestamp, Gate_ID, Override_Reason[cite: 1].	Enforces a strict compound unique constraint on (Event_ID, Student_ID) for active tickets[cite: 1]. Self-cancellation is permitted only during active registration windows prior to gate scanning[cite: 1].

11. OPERATIONAL EDGE CASE RESOLUTION MATRIX

Operational Scenario	Triggering State / Event	Systemic Resolution & Enforcement Rule
Cancellation Attempt Post-Window [cite: 1]	REGISTRATION_CLOSED[cite: 1]	Self-cancellation button is permanently hidden and disabled on the dashboard[cite: 1]. Registered seats remain locked to preserve operational roster integrity[cite: 1].
The Cancellation Bounce [cite: 1]	FULLY_BOOKED during active registration window[cite: 1]	A student cancellation decrements booked count from N to $N - 1$ [cite: 1]. System immediately flips status to OPEN and enables public registration[cite: 1].

Operational Scenario	Triggering State / Event	Systemic Resolution & Enforcement Rule
Cancellation Under Manual Lock [cite: 1]	MANUALLY_LOCKED active override[cite: 1]	Count decrements in administrative roster table, but public status remains locked[cite: 1]. Dynamic reopening bounce is suppressed[cite: 1].
Scan of a Cancelled Ticket [cite: 1]	Ticket marked REVOKED[cite: 1]	Gate terminal displays [AMBER] REVOKED TICKET ALERT, displays cancellation timestamp, and blocks entrance[cite: 1].
Duplicate Pass-Back Scan Attempt [cite: 1]	Ticket marked CHECKED_IN[cite: 1]	Gate terminal displays [RED] DUPLICATE ADMISSION ALERT, displaying initial check-in timestamp and gate terminal identifier[cite: 1].
Admin Scan Rollback [cite: 1]	Accidental check-in scan[cite: 1]	Rollback restores ticket status to <i>Pending Arrival</i> and decrements headcount gauge, while keeping student self-cancellation locked[cite: 1].
Schedule or Venue Relocation [cite: 1]	Administrative update to active event[cite: 1]	Saving modified event logistics triggers automated email updates to all confirmed ticket holders[cite: 1].
Event Cancellation [cite: 1]	CANCELLED triggered by administrator[cite: 1]	Prompts for mandatory reason, voids all tickets, and dispatches batch cancellation emails to all registrants[cite: 1].